

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and enhanced **ML compilers** using **PyTorch** and **C++** for a specialized Neural Processing Unit (NPU) architecture, optimizing critical code paths.
- Designed and implemented advanced **quantization techniques** to improve model efficiency and accuracy for hardware deployment, reducing model size by an average of **38%**.
- Optimized **ML workloads** for deployment on embedded hardware, ensuring top-tier performance and energy efficiency, measured by a **15% reduction** in inference cycles.
- Collaborated with internal architecture and software engineering teams to integrate novel **ML compiler technologies** into core platforms.
- Researched and prototyped innovative solutions for **system-level optimizations** in ML compilation, contributing to internal technical whitepapers.
- Evaluated and debugged **ML performance** across complex workflows, identifying and resolving computational bottlenecks that impacted runtime by up to **20%**.

SKILLS

ML Frameworks – PyTorch, TensorFlow, ONNX

Programming Languages – C++, Python

ML Concepts – Compiler Development, Quantization, ML Workload Analysis, Performance Optimization, System-Level Optimization, Hardware-Aware ML

Tools & Technologies – ASIC Design, Verification, Validation, Integration, Git, Jenkins

Cloud & Data – AWS, PostgreSQL, MySQL, Oracle

PROJECTS

NPU Compiler Optimization for PyTorch Models

- Developed a C++ module for an ML compiler framework to better optimize PyTorch model operations for NPU execution, focusing on tensor manipulation and memory access patterns.
- Implemented a new quantization pass that reduced model footprint by **~35%** while maintaining over **90%** of original accuracy on benchmark datasets.
- Integrated and tested the compiler modifications using a suite of standard ML workloads, analyzing performance regressions and improvements using custom profiling tools.

Hardware-Aware ML Model Efficiency Tuning

- Engineered Python scripts to analyze and tune ML models (PyTorch) for deployment on resource-constrained hardware, focusing on reducing computational complexity and power consumption.
- Applied various pruning and quantization strategies, achieving a **22% reduction** in FLOPS and a **40% decrease** in memory bandwidth requirements for a computer vision model.
- Benchmarked optimized models against baseline versions on simulated hardware profiles, documenting performance gains and potential trade-offs in accuracy.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)